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Inherited Disorders of the Glomerulus

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CHAPTER OUTLINE

INTRODUCTION

Over the past 2 decades, there has been a rapid growth in our understanding of the influence of genetics in the development of glomerular disease. Although individual mendelian diseases make up only a small fraction of glomerular pathology,[1] it has become apparent that in aggregate they may represent a significant portion of disease in select populations such as pediatric steroid-resistant nephrotic syndrome.[2–4] The discovery that common risk variants in the *APOL1* gene are a major contributor to the increased risk of focal segmental glomerulosclerosis (FSGS), HIV-associated nephropathy (HIVAN), and nondiabetic end-stage kidney disease (ESKD) in populations of West African descent has dramatically altered our perspective of this health disparity.[5,6] Common genetic variants have been associated with increased risk of membranous nephropathy,[7,8] immunoglobulin A (IgA) nephritis,[9–12] and steroid-sensitive nephrotic syndrome.[13,14] At a basic level, genetic forms of glomerular disease have provided significant insight into the key structures necessary for the proper function and maintenance of the glomerular filtration barrier, with a particular focus on the glomerular podocyte and basement membrane. Clinically, the identification of genetic forms of glomerular disease has begun to have an impact on therapeutic considerations, risk assessment for disease recurrence posttransplantation, and counseling of potentially affected family members and kidney donors. As such, genetic factors are primed to affect the development of personalized medicine for glomerular disease.

Broadly speaking, genetic diseases are typically divided into mendelian disorders and polygenic disorders. In mendelian diseases, single-gene mutations give rise to highly penetrant pathologic phenotypes. With few exceptions, these disorders are further classified into dominant and recessive diseases, depending on whether a single mutant allele is sufficient to cause disease (dominant) or whether the lack of a normal, wild-type allele is the driver of disease (recessive). A detailed family history is often informative in suggesting the pattern of disease inheritance; dominant diseases classically affect members of multiple generations, whereas recessive disease is clustered among members of a single generation. However, it is important to realize that mendelian diseases can appear to be sporadic in the setting of small families or incompletely penetrant phenotypes and should not be ruled out by the absence of a family history. In contrast to mendelian diseases, multiple genetic variants or mutations influence the development of polygenic disorders. In these cases, individual variations are often seen in a significant proportion of the general population, but have a small effect size in regard to altering disease risk. Thus, these genetic variants are often of interest in terms of population risks and understanding the mechanism of disease, but are rarely clinically actionable.

The last 20 years have seen an explosion in the number of genes implicated in mendelian forms of glomerular disease. Mutations in over 50 genes have been reported to associate with various glomerular pathologies, including those associated with multiorgan syndromic phenotypes (Table 43.1). Studies of the gene products have identified several structural and functional components of the glomerulus that appear particularly critical for normal function or vulnerable to failure. Specifically, multiple genetic mutations target components of the glomerular basement membrane and several subcellular structures of the podocyte (Fig. 43.1).

DISORDERS OF THE PODOCYTE

Podocytes, or visceral epithelial cells of the glomerulus, represent the outer layer of the glomerular filtration barrier. Derived from the tip of the developing nephron, these cells

Table 43.1 Genes Involved in Inherited Human Glomerular Diseases

Gene	Mode of Inheritance	Protein	Site of Action	Extraglomerular Manifestations	OMIM No.
NPHS1	AR	Nephrin	SD		256300
NPHS2	AR	Podocin	SD		600995
CD2AP	AR, AD	CD2-associated protein	SD		607832
TRPC6	AD	Transient receptor potential C6	SD		603965
MYO1E	AR	Nonmuscle myosin IE	SD, AC		614131
FAT1	AR	FAT atypical cadherin 1	SD	Renal tubular ectasia	*600976
MAGI2	AR	Membrane-associated guanylate kinase, WW, and PDZ domain-containing 2	SD, AC	Neurologic deficits	617609
ACTN4	AD	α-Actinin-4	AC		603278
INF2	AD	Inverted formin 2	AC, mito (?)	Charcot-Marie-Tooth	613237, 614455
MYH9	AD	Nonmuscle myosin heavy chain-9	AC	Fechtner syndrome	153640
ARHGAP24	AD	Rho GTPase activating protein 24	AC		*610586
ARHGDIA	AR	Rho GDP dissociation inhibitor 1	AC		615244
ANLN	AD	Anilin	AC		616032
AVIL	AR	Advillin	AC		*613397
KANK1	AR	KN motif and ankyrin repeat domains 1	AC	Intellectual disability	* 607704
KANK2	AR	KN motif and ankyrin repeat domains 2	AC		* 614610
KANK4	AR	KN motif and ankyrin repeat domains 4	AC	Intellectual disability, facial dysmorphism, ASD	* 614612
TNS2	AR	Tensin-2	AC		607717
DLC1	AR	Deleted in liver cancer 1; ARHGAP7	AC		604258
ITSN1	AR	Intersectin 1	AC		602442
ITSN2	AR	Intersectin 2; SH3D1B	AC		604464
CDK20	AR	Cyclin-dependent kinase 20		MPGN	610076
COL4A3,4,5	XLR, AR, AD	Collagen 4 α_3, α_4, α_5	GBM	Alport syndrome, deafness	104200, 203780, 301050
LAMB2	AR	Laminin β_2	GBM	Pierson syndrome	609049
ITGA3	AR	Integrin α_3	GBM adhesion	Interstitial lung disease, epidermolysis bullosa	614748
CD151	AR	CD151 (Raphe blood group)	GBM adhesion	Pretibial epidermolysis bullosa, deafness	609057
COQ2	AR	Coenzyme Q2 4-hydroxybenzoate polyprenyltransferase	CoQ10; mito (?)	Encephalopathy, cardiomyopathy	607426
PDSS2	AR	Prenyl (decaprenyl) diphosphate synthase, subunit 2	CoQ10, mito (?)	Leigh syndrome	614652
COQ6	AR	Coenzyme Q6 monooxygenase	CoQ10, mito (?)	Deafness	614650
ADCK4	AR	aarF domain containing kinase 4	CoQ10, mito (?)		615573
MT-TL1	Maternal	Mitochondrially-encoded tRNA leucine 1 (UUA/G)	Mito	MELAS syndrome, diabetes, deafness	590050
WT1	AD	Wilms' tumor 1	Gene transcription	Denys-Drash syndrome, Frasier syndrome	194080, 136680
LMX1B	AD	LIM homeobox transcription factor 1β	Gene transcription	Nail-patella syndrome	161200
SMARCAL1	AR	SWI/SNF–related, matrix-associated, actin-dependent regulator of chromatin, subfamily A-like protein 1	Gene transcription	Schimke immunoosseous dysplasia	242900
PAX2	AD	Paired box gene 2	Gene transcription	Papillorenal syndrome	120330
NXF5	XLR	Nuclear RNA export factor 5	Nuclear export	Heart block	300319
NUP93	AR	Nucleoporin, 93 kDa	Nuclear pore		616892
NUP205	AR	Nucleoporin, 205 kDa	Nuclear pore		616893
XPO5	AR	Exportin 5	Nuclear export		607845
NUP107	AR	Nucleoporin, 107 kDa	Nuclear pore	Galloway-Mowat syndrome	616730
WDR73	AR	WD40 repeat-containing protein 73	Unknown	Galloway-Mowat syndrome	251300
OSGEP	AR	O-sialoglycoprotein endopeptidase	KEOPS complex	Galloway-Mowat syndrome	251300

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Table 43.1 Genes Involved in Inherited Human Glomerular Diseases (Cont'd)

Gene	Mode of Inheritance	Protein	Site of Action	Extraglomerular Manifestations	OMIM No.
TP53RK	AR	TP53 regulating kinase	KEOPS complex	Galloway-Mowat syndrome	251300
TPRKB	AR	TP53RK binding protein	KEOPS complex	Galloway-Mowat syndrome	251300
LAGE3	AR	L antigen family member 3	KEOPS complex	Galloway-Mowat syndrome	251300
SCARB2	AR	Scavenger receptor class B, member 2	Lysosome	Action myoclonus –renal failure syndrome	254900
PMM2	AR	Phosphomannomutase-2	Protein glycosylation	Congenital disorder of glycosylation type Ia	212065
ALG1	AR	β-1,4-mannosyltransferase	Protein glycosylation	Congenital disorder of glycosylation type Ik	608540
SLC35A2	AR	UDP-galactose transporter	Protein glycosylation	Congenital disorder of glycosylation type IIm	300896
EMP2	AR	Epithelial membrane protein 2	Unknown		615861
APOL1	AR	Apolipoprotein L1	Unknown		612551
PLCE1	AR	Phospholipase C ε1	Unknown, SD (?)		610725
DGKE	AR	Diacylglycerol kinase ε	Unknown	Atypical HUS	615008
SGPL1	AR	Sphingosine-1-phosphate lyase	Sphingolipid catabolism	Ichthyosis, acanthosis, adrenal insufficiency, immunodeficiency, neuronal dysfunction	617575
CRB2	AR	Crumbs homolog 2	Apical-basal cell polarity	Ventriculomegaly with cystic kidney disease	616220, 219730
PTPRO	AR	Glomerular epithelial protein-1	Apical membrane		614196
PODXL	AD	Podocalyxin	Apical membrane		*602632
CUBN	AR	Cubilin	Proximal tubule	Megaloblastic anemia	261100
FN1	AD	Fibronectin 1	Fibronectin deposits		601894

AC, Actin cytoskeleton; *AD,* autosomal dominant; *AR,* autosomal recessive; *CoQ10,* coenzyme Q10 synthesis; *GBM*, glomerular basement membrane; *HUS,* hemolytic-uremic syndrome; *MELAS,* mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes; *mito,* mitochondria; *OMIM,* Online mendelian Inheritance in Man; *SD,* slit diaphragm; *XLR,* X-linked recessive; *UDP,* uridine diphosphate.

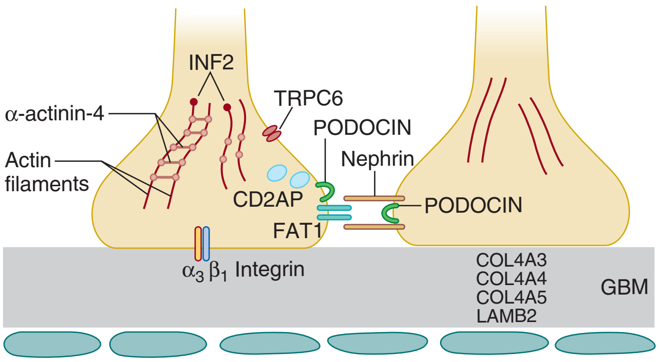

Fig. 43.1 Schematic depiction of the glomerular filtration barrier. The three major components of the filtration barrier—the interdigitated podocyte foot processes, the glomerular basement membrane *(GBM)*, and the fenestrated capillary endothelium—are depicted. Mendelian forms of glomerular disease affect multiple proteins localized to distinct substructures of the filtration barrier, suggesting that these are potential disease nodes critical to normal function. These include components of the GBM and GBM-podocyte adhesion complexes, the slit diaphragm, a specialized cell-cell junction spanning adjacent foot processes, and the actin cytoskeleton found within foot processes, all shown here. Additional cellular structures repeatedly targeted by disease gene mutations, but not shown here, include mitochondria and the nuclear pore. In the interest of clarity, not all slit diaphragm and actin cytoskeleton components and regulators whose mutation cause human disease are depicted. A more comprehensive list of these genes and their proposed site of action, if known, can be found in Table 43.1.

form a continuous epithelial sheet lining the urinary side of the glomerular basement membrane. Podocytes are characterized by several key features:

1. A highly arborized architecture, with foot processes of adjacent podocytes forming an intricate interdigitating array dependent on the cells' actin cytoskeleton
2. The slit diaphragm, a specialized cell-cell contact spanning adjacent foot processes that has been proposed to act as the final component of the highly selective glomerular filtration barrier
3. Exposure to the large hydrodynamic pressure responsible for transglomerular filtration
4. A highly differentiated state with very limited, if any, replicative capacity

Human genetics, bolstered by in vitro and animal studies, have been instrumental in establishing several disease nodes in the podocyte.

The highly arborized architecture of podocytes is frequently disrupted, a process termed "foot process effacement," in proteinuric kidney diseases. Effacement usually also disrupts the slit diaphragm. Normally, the slit diaphragm appears as a thin, zipper-like structure by electron microscopy[15] and has been compared with a modified adherens and tight junction,[16] with unique molecular components. Passage of the primary glomerular filtrate is thought to occur through the slit diaphragm, which has been proposed to act as the final component of the highly selective glomerular filtration barrier. Human genetic studies were instrumental in establishing the importance of the slit diaphragm for proper podocyte, and hence glomerular, function when *NPHS1* and *NPHS2* genes responsible for congenital nephrotic syndrome and a form of autosomal recessive nephrotic syndrome, respectively, were found to encode components of the slit diaphragm.[17,18] Mutations in several additional slit diaphragm genes have since been identified, which, when mutated, cause nephrotic syndrome and/or FSGS. Work over the subsequent 2 decades has identified a large number of rare mendelian forms of nephrotic syndrome and FSGS, as well as common genetic variants that influence the risk of various forms of glomerular injury.

MENDELIAN DISEASES OF THE PODOCYTE

These include congenital nephrotic syndromes and other forms of monogenic, steroid-resistant nephrotic syndrome.

CONGENITAL NEPHROTIC SYNDROMES

NEPHRIN *(NPHS1)*

Mutations in the nephrin gene, *NPHS1*, cause a severe, neonatal onset form of nephrotic syndrome, referred to as "congenital nephrotic syndrome," or CNS.[18] *NPHS1* was identified through traditional positional cloning methods, in which genetic linkage analysis leads to the identification of a genomic locus containing a gene alteration segregating with disease and, ultimately, identification of the gene itself. Nephrin is a 1,241–amino acid transmembrane protein that is a member of the immunoglobulin family of cell adhesion molecules. Two founder mutations in the Finnish population—$\text{Fin}_{\text{major}}$ and $\text{Fin}_{\text{minor}}$—account for most of the disease-causing NPHS1 mutations in Finland. $\text{Fin}_{\text{major}}$ is a 2-base pair (bp) deletion early in the NPHS1 coding sequence that causes a frameshift, leading to an early stop codon. The $\text{Fin}_{\text{minor}}$ allele encodes a premature stop codon (R1109X). A founder effect explains the high rate of these two mutations and CNS in Finland. Although CNS is seen in about 1 in 8200 live births in Finland, it is much rarer outside of Finland (~1–3/100,000 births).[19]

NPHS1 mutations occur worldwide. Over 200 distinct disease-associated mutations have been reported.[20,21] These appear to be biologically loss-of-function mutations. Most disease-associated mutations are missense but other types of mutations have also been reported (e.g., splice mutations, early termination, frameshifts).[22] Although *NPHS1* mutations are perhaps the most common cause of CNS, they are not the sole cause. *NPHS1* mutations account for virtually all cases of CNS in Finland, but from 40% to 80% of cases in other reports.[18,23,24] In addition to Finland, where CNS is more common than in other countries, founder mutations have been reported elsewhere. For example, a founder mutation in population has been described in Old Order Mennonites.[25] The large spectrum of mutations, including clear loss-of-function alleles and the knockout mouse phenotype, support the notion that this is a typical loss-of-function disorder.[26]

Nephrin localizes to the podocyte slit diaphragm.[27,28] Nephrin molecules perform structural and signaling functions.[29] Nephrin molecules on adjacent foot processes form a zipper-like structure that functions as an essential component of the glomerular filtration apparatus.[27,28] The intracellular domain of nephrin interacts with multiple intracellular signaling pathways. Nephrin interacts directly with the SH2/SH3-domain containing Nck proteins to mediate nephrin-dependent actin cytoskeleton dynamics.[30,31] Interactions between nephrin and other cytoskeletal proteins, including CD2AP, connect the nephrin signaling complex with other actin regulating proteins.[32] Nephrin and CD2AP interact with PI3-kinase (PI3K), recruit it to the plasma membrane and, with podocin, stimulate PI3K-dependent AKT signaling.[33]

Nephrin-mediated CNS is a severe disease and is lethal when untreated because of the consequences of severe nephrosis or ESKD.[18,34] Affected neonates can excrete on the order of 20 to 30 g of protein in the urine/day and demonstrate other manifestations of severe nephrosis—hypercoagulability, hypoalbuminemia, and edema.[19] Kidneys are typically somewhat enlarged. Kidney histology typically shows a normal glomerular basement membrane appearance but diffuse podocyte foot process effacement and dilation of the proximal tubules.[35] Initial supportive treatment may include the use of diuretics to control edema and albumin infusions for hypoalbuminemia. Aggressive management also includes bilateral nephrectomy to prevent nephrosis and its complications and the initiation of renal replacement therapy with peritoneal dialysis, ultimately to be followed by kidney transplantation when the affected child has reached an adequate size, typically at about 1 year of age.[36,37] Alternative treatment plans have been suggested, including a regimen including unilateral nephrectomy, renin-angiotensin system blockade, and nonsteroidal antiinflammatory drugs (NSAIDs) to reduce proteinuria and the need for albumin infusion, with later initiation of renal replacement therapy.[38] Kidney transplantation is considered standard therapy for nephrin-mediated CNS. Although nephrin-mutant CNS itself does not recur in the allograft, proteinuria secondary to the development of antinephrin antibodies is commonly observed.[39] Alpha-fetoprotein (AFP) levels in the maternal serum and amniotic fluid are elevated in CNS.[40] In a family known to be at high risk for CNS (e.g., because of Finnish parents or a previous child with CNS), prenatal genetic testing can be performed.

PODOCIN *(NPHS2)*

Mutations in the podocin gene *NPHS2* cause a spectrum of inherited disorders of the podocyte, ranging from neonatal nephrotic syndrome to late-onset FSGS.[17,41] In most of the reported cases, steroid-resistant nephrotic syndrome (SRNS) and FSGS associated with *NPHS2* mutations present in early childhood. *NPHS2*-associated disease is resistant to treatment with corticosteroids.[42] Causal mutations in *NPHS2* have been reported to occur in a high fraction of children with SRNS, as high as 30% in some studies[43,44]

The clinical phenotype associated with *NPHS2* mutations is highly variable. In some reports, *NPHS2* mutations are as common a cause of CNS as nephrin mutations.[24] At the other end of the clinical spectrum, *NPHS2* mutations are also associated with adult-onset FSGS, also under a recessive model of disease inheritance. In particular, a relatively common coding sequence variant in *NPHS2, R229Q*, is associated with late-onset FSGS when inherited with a second, more severe mutation.[45,46] Interestingly, the pathogenicity of the *R229Q* variant depends on the specific second allele and leads to overt disease only when inherited together with certain C-terminal *NPHS2* mutations by causing altered heterodimerization and mislocalization.[47]

Well over 100 distinct disease-associated *NPHS2* mutations have now been reported. Such mutations have been observed in all regions of the gene and include point mutations, early termination mutations (frameshift and early stop codons), and insertions and deletions.[41,48,49] An *R138Q* mutation has been reported in many apparently unrelated individuals and families and is typically associated with a severe phenotype.[50]

Podocin, a 383–amino acid integral membrane protein, localizes to the glomerular slit diaphragm.[51] Podocin traffics to the plasma membrane through the classic exocytic pathway. A direct interaction between podocin and the tyrosine-phosphorylated cytoplasmic domain of nephrin appears to facilitate nephrin-mediated signaling.[29] Podocin can also influence TRPC6 channel activity.[52,53] Most disease-causing podocin mutations lead to retention in the endoplasmic reticulum (ER).[54] Mutants that are retained in the ER appear to be associated with more severe disease than those that traffic to the cell membrane. Presumably, disease results from a consequent defect in slit diaphragm structure. Podocin may function as a dimer or oligomer via C-terminal interactions. In the presence of two *NPHS2* mutations, the specific nature of these mutations may affect the oligomerization and nature of the associated phenotype.[55] Mouse models are consistent with the human genetic findings. Mice lacking podocin demonstrate severe kidney disease and altered glomerular filtration.[56] Podocin is important in maintaining the adult filtration apparatus; mice engineered to allow podocin inactivation in adult mouse kidneys show FSGS and nephrotic syndrome.[57]

A genetic diagnosis of *NPHS2*-mediated disease can be clinically actionable. Although some immunosuppressive therapies may mediate effects on podocytes through nonimmunologic mechanisms, aggressive immunosuppressive medication is probably best avoided in individuals with *NPHS2*-associated disease.[42] Treatment of *NPHS2*-associated disease is largely supportive. Therapy with angiotensin receptor blockers (ARBs) or angiotensin-converting enzyme (ACE) inhibitors is standard. *NPHS2* mutations are not a contraindication to kidney transplantation; on the contrary, SRNS and FSGS do not tend to recur in such patients after transplantation,[4] although there are rare exceptions.[50,58] This knowledge may affect posttransplantation care.

ACTN4

Mutations in the alpha-actinin 4 gene *ACTN4* cause an autosomal dominant form of kidney disease characterized by late-onset proteinuria, chronic kidney disease (CKD), and a slow progression toward kidney failure.[59] Alpha-actinin 4 is a widely expressed homodimeric actin binding and cross-linking protein. All mutations that have been convincingly shown to cause human disease are located in the actin-binding domain (ABD) of the encoded protein.[60] These mutations alter the interaction of alpha-actinin 4 with actin filaments. Despite the widespread expression of the alpha-actinin 4 protein, disease manifestations have only been observed in the glomerulus. Although *ACTN4*-associated disease is typically late in onset and usually presents with subnephrotic proteinuria, this is not always the case. Childhood-onset forms of alpha-actinin 4–associated disease have been reported.[61,62] Although clinical data are sparse, it appears that patients with *ACTN4*-mediated FSGS are resistant to immunosuppressive therapy, as might be anticipated in the setting of an altered cytoskeletal protein. Similarly, recurrence of FSGS in transplanted kidneys has not been reported in these patients.

Mice lacking *Actn4,* as well as mice with point-mutant forms of *ACTN4,* demonstrate glomerular disease.[63–65] Disease-causing ACTN4 mutations alter actin dynamics and the biophysical properties of the actin cytoskeleton.[65–67] Cells harboring disease-associated *ACTN4* mutations are stiffer and more susceptible to breakage than wild-type cells. Some studies have suggested noncytoskeletal effects of ACTN4 as well. ACTN4 has been shown to potentiate nuclear factor-kappa B (NF-κB)activity in a manner that is independent of its cytoplasmic actin-binding function.[68] Through this nuclear role, ACTN4 has been shown to regulate glucocorticoid receptor–mediated gene regulation in podocytes.[69]

INF2

Mutations in the *INF2* gene cause a late-onset form of kidney disease that is clinically similar to that associated with *ACTN4* mutations.[70] INF2 is a member of the family of mammalian formin proteins. INF2, like other formins, is an actin regulatory protein. INF2 forms a homodimer that associates with the barbed end of an actin filament, promoting actin polymerization and depolymerization.[71] An FH2 domain near the C-terminus of INF2 is responsible for the actin-polymerizing activity. An interaction between a C-terminal “diaphanous autoregulatory domain” (DAD) and an N-terminal “diaphanous inhibitory domain” (DID) normally keeps INF2 in an inhibited state.[71,72] All the *INF2* mutations associated with human FSGS reported to date localize to this N-terminal region. The penetrance of *INF2*-mediated disease appears to be lower than that for *ACTN4*- and *TRPC6*-associated autosomal dominant FSGS. Mutations in *INF2* appear to be among the most common forms of podocyte-mediated autosomal dominant FSGS, with over 45 different disease-associated mutations reported to date.[70,73–76]

A subset of patients with *INF2*-associated FSGS also have Charcot-Marie-Tooth (CMT) disease.[75] CMT is a peripheral demyelinating neuropathy with several genetic causes. It is not yet clear why some *INF2* mutations cause FSGS alone and others cause FSGS plus CMT. Disease-associated *INF2* mutations appear to both inhibit the ability of the DID region to interact with the DAD, as well as the ability of the DID to interact with other formin family members.[72]

TRPC6

Mutations in the *TRPC6* gene cause an autosomal dominant kidney disease characterized by proteinuria, focal and segmental glomerulosclerosis, and progressive renal failure.[77] TRPC6 is a member of the canonical (or TRPC) subfamily of transient receptor potential (TRP) channels.[78] It forms cation channels as either homotetramers or heterotetramers in conjunction with TRPC1, TRPC3, and TRPC7 and is activated by diacylglycerol; its activation leads to elevations in intracellular calcium levels.[79,80] TRPC6 is expressed in podocytes and reported to interact with nephrin and podocin, with both affecting TRPC6 function.[52,53,81,82] In rodents, TRPC6 modulates pathologic responses downstream of angiotensin

II (Ang II) signaling, Gαq activation, and several other disease models.[83–87] Numerous potential pathways downstream of TRPC6 activation have been reported,[87–93] although their relative importance in the pathogenesis of genetic forms of TRPC6-associated disease remain uncertain.

Both gain-of-function[77,81,94] and loss-of-function mutations[95] in TRPC6 have been reported to cause glomerular disease. TRPC6-mediated disease is incompletely penetrant; it predominantly becomes apparent in adults, although pediatric cases have been reported.[77,81,94,96,97] Although calcineurin inhibitors have been hypothesized as a potential treatment, this has not been reported effective in practice.[98]

OTHER FORMS OF MONOGENIC STEROID-RESISTANT NEPHROTIC SYNDROME

A large number of other genes have been identified that when mutated, cause recessive SRNS and/or FSGS. These can be roughly categorized by the biologic pathway disrupted by mutations. The vast majority of these are early-onset recessive diseases. Some have extrarenal manifestations, and some do not.

CYOSKELETON

Actin-Binding Proteins

These are in addition to *INF2, ACTN4, MYH9, MYO1E,* and *ANLN*. Mutations in the nonmuscle type II myosin gene *MYH9* cause a set of autosomal dominant disorders characterized by leukocyte inclusions, platelet abnormalities, hearing loss, and nephropathy.[99] Although different constellations of *MYH9*-associated manifestations have been given the names May-Hegglin anomaly, Sebastian syndrome, Fechtner syndrome, and Epstein syndrome, these are probably all best considered as part of a spectrum of diseases of altered *MYH9*. Approximately 30% of patients with disease-associated *MYH9* mutations have manifestations of kidney disease.[100] Kidney histology shows an FSGS-like lesion. There are correlations between mutation and phenotype. *MYH9* mutations that alter the motor domain of the encoded myosin typically show severe thrombocytopenia and deafness before 40 years of age.[101] Patients with motor domain mutations are also more likely to have severe kidney disease and progression to kidney failure.[102] Amino acid residue 702 appears to be a hot spot for mutations leading to severe disease, including a severe nephropathy.[103]

Mutations in *MYO1E* cause a rare, childhood-onset, steroid-resistant form of FSGS.[104,105] *MYO1E* encodes an atypical type 1 myosin, myosin 1E, that has been implicated in cell adhesion, migration, and endocytosis.[106] Mice deficient in myosin 1E demonstrate glomerular disease.[107]

Mutations in the gene encoding the actin-binding protein anilin *(ANLN)* have been reported as a rare cause of autosomal dominant FSGS. To date, two disease-associated point mutations have been reported. Mutations are hypothesized to disrupt normal regulation of the slit diaphragm–associated actin cytoskeleton.[108]

Mutations in the actin-associated protein advillin (*AVIL*) cause a recessive form of SRNS.[109] These mutations alter the actin-binding activity of the protein. Because disease-associated *AVIL* mutations were found to alter the association of phospholipase C epsilon with the ARP2/3 complex, *AVIL* has been hypothesized to act upstream of this enzyme to control ARP2/3-dependent actin regulation in the podocyte.

Cytoskeletal Regulators

Mutations in members of the kidney ankyrin repeat-containing protein family (*KANK1, KANK2, KANK4*) have been reported to cause early childhood-onset SRNS under a recessive model.[110] Individuals with *KANK4* mutations were also reported to have nonrenal features, including facial dysmorphism. These mutations are hypothesized to alter Rho GTPase signaling. Mutations in the Rho GDI-alpha gene *ARHGDIA* are a rare cause of childhood-onset recessive SRNS thought to perturb the same pathway.[111,112] The gene product is a regulator of Rho GTPase activity. Disease-associated mutations lead to altered interactions with Rho GTPase family members, as well as altered activity of some of these proteins. Mice homozygous for an inactivated *Arhgdia* gene show severe proteinuria and develop kidney failure.[113]

ARHGAP24 encodes a RhoA-activated Rac1 GTPase-activating protein, also known as FilGAP.[114] Like other GTPase-activating proteins, ARHGAP24 regulates the activity of the Rho family members Cdc42, Rac, and RhoA. In podocytes, ARHGAP24 inactivates Rac1 activity.[115] A disease-segregating point mutation that abrogates GAP activity has been reported in a family with FSGS.[115] Mutations in *MAGI2, TNS2, DLC1, CDK20, ITSN1,* and *ITSN2* have been reported to cause rare forms of autosomal recessive, partially SRNS in children.[116] Alteration of small GTPase activity has been proposed as the unifying feature of these forms of nephrotic syndrome.

SLIT DIAPHRAGM

In addition to *NPHS1* (nephrin) and *NPHS2* (podocin), mutations in other slit diaphragm– associated proteins have been reported as rare causes of early-onset nephrotic syndrome.

CD2AP

CD2-associated protein (CD2AP) is a slit diaphragm–associated scaffolding molecule involved in actin cytoskeleton regulation. CD2AP was identified on the basis of its interaction with the T cell adhesion molecule CD2.[117] Mice lacking CD2AP were found to have severe proteinuria and the development of glomerulosclerosis.[118] In the podocyte, CD2AP colocalizes with and binds to the nephrin C-terminus at lipid rafts.[119,120] CD2AP and nephrin both interact with the p85 subunit of PI3K and stimulate PI3K-dependent AKT signaling in podocytes.[33] The absence of CD2AP increases transforming growth factor beta 1 (TGF-β1)–induced activity of the proapoptotic p38 MAPK pathway.[121] Heterozygosity for loss-of-function mutations in CD2AP appears to be a rare cause of, or contributor to, human FSGS.[122] A case of apparent recessive CD2AP-associated FSGS has been reported in a 10- month-old child.[123] The small number of CD2AP-associated cases of FSGS nephrotic syndrome make it difficult to generalize as to mode of inheritance or disease mechanism.

FAT1

Mutations in the *FAT1* gene, encoding a protocadherin family member localizing to the slit diaphragm, have been reported as a cause of glomerular and tubular ectasia, as well as neurologic disease, in several families.[124]

Similarly, mice lacking FAT1 have a lethal phenotype characterized by loss of slit diaphragms and severe defects in brain development.[125]

MAGI-2

Membrane-associated guanylate kinase inverted 2 (MAGI-2) is a large scaffolding protein present at the podocyte slit diaphragm, where it binds nephrin and regulates cytoskeletal dynamics.[126] Its transcription is regulated by Wilms' tumor 1.[127,128] MAGI-2 mutations have been reported as a rare case of autosomal recessive congenital nephrotic syndrome in children.[116,129] Mice lacking MAGI-2 similarly develop severe glomerulosclerosis, with impaired slit diaphragm formation and podocyte loss.[130–132]

PODOCYTE-MATRIX ADHESION

Several additional gene mutations have been reported to cause FSGS or nephrotic syndrome by altering the interaction of podocytes with the glomerular basement membrane (GBM). Mutations in the integrin beta 4 gene (*ITGB4*) have been reported to cause a recessive form of epidermolysis bullosa and congenital nephrosis.[133] A recessive mutation in integrin alpha 3 (*ITGA3*) causing a gain of a glycosylation site has been reported to lead to congenital nephrosis presenting together with interstitial lung disease.[134] Mutations in the *CD151* gene (cluster of differentiation 151), encoding a tetraspanin protein that modulates integrin-laminin interactions,[135] have been reported to cause another syndrome of proteinuric kidney disease, epidermolysis bullosa and sensorineural deafness.[136] CD151-deficient mice similarly develop FSGS and kidney failure.[137] Mutations in the exostosin gene *EXT1* cause multiple exostoses type I, an autosomal dominant bone disorder.[138] EXT1 is a transmembrane glycoprotein important for the synthesis and display of cell surface heparan sulfate glycosaminoglycans.[139] A family has been reported in which steroid-sensitive nephrotic syndrome, in addition to multiple exostoses, segregated with an *EXT1* mutation.[140] The Alport syndrome–associated type 4 collagen genes and Pierson syndrome disease gene *LAMB2* are discussed later.

MITOCHONDRIA

Alterations in mitochondria lead to a diverse set of phenotypic manifestations.[141] Inherited mitochondrial disorders are often divided into disorders of the mitochondrial genome and disorders caused by mutations in nuclear genes that encode proteins important for mitochondrial function. Mutations in the mitochondrial genome, in particular mtDNA 3243 A to G, lead to a syndrome called MELAS (*m*itochondrial myopathy, *e*ncephalopathy, *l*actic *a*cidosis and *s*trokelike episodes). Kidney disease is not a particularly common feature, but may be seen, and can present as proteinuria with FSGS histology.[141] The maternally derived inheritance and variability in phenotype make this a difficult diagnosis.

Defects in coenzyme Q10 (CoQ10) biosynthesis have been recognized as a cause of glomerulopathy, either isolated or part of a multisystem syndrome.[142] Multiple enzymatic steps are required for CoQ10 synthesis, and mutations in several of the genes encoding critical enzymes have been found to be mutated in individuals with SRNS.[143–148] Typically, affected individuals present in childhood and demonstrate proteinuria and FSGS-like histology. Although cases of isolated nephropathy have been reported, nonrenal features are common. Neurologic features, often severe, may be the predominant clinical features of disease. Mutations in *COQ2,* encoding an enzyme early in the CoQ10 synthesis pathway, were the first genetic cause of CoQ10 deficiency identified.[143] *COQ2*-related disease follows recessive inheritance, as is typical for a disease caused by an enzyme deficiency. Multiple *COQ2* mutations have since been identified in unrelated individuals.[144]

Other causes of SRNS related to defects in CoQ10 synthesis include mutations in *ADCK4, PDSS2,* and *COQ6*.[145–148] Importantly, disease features, including the glomerulopathy, may respond to CoQ10 supplementation.[149,150]

NUCLEAR PORE PROTEINS

Mutations in nuclear pore genes *NUP93, NUP107, NUP205,* and *XPO5* all cause rare forms of recessive, early-onset SRNS.[151–154] Clinically, patients can also present with microcephaly, similar to Galloway-Mowat syndrome. The mechanism connecting defects in these pore proteins with nephrotic syndrome are not understood, although aberrant SMAD signaling has been proposed.[152]

TRANSCRIPTION FACTORS

LMX1B

Nail-patella syndrome is caused by mutations in the LMX1B transcription factor.[155,156] This disorder follows autosomal dominant transmission. The nonrenal phenotype is characterized by small or absent patellae and absent or dysplastic fingernails and toenails. It is estimated that a glomerular defect is present in 40% of cases of nail-patella syndrome.[157] The kidney phenotype is highly variable. Some individuals develop early-onset nephrotic syndrome, whereas others develop much milder CKD.[158] Typically, kidney histology shows FSGS and a thickened GBM.[159] Although the skeletal manifestations are present in most patients, families with apparent LMX1B-associated kidney disease but lacking skeletal manifestations have been described.[160–162]

LMX1B is a transcription factor that plays a crucial role in the transcriptional regulation of essential podocyte genes.[163,164] Studies of the activity of point mutant LMX1B have similarly supported the hypothesis that this is a loss-of-function disease.[165,166] Multiple families have been identified in which the genetic lesion appears to be a deletion of the entire *LMX1B* gene, suggesting that haploinsufficiency, rather than a gain-of-function mechanism, is responsible for this autosomal dominant disease.[167]

WT1

Mutations in the Wilms' tumor gene, the tumor suppressor *WT1,* cause a spectrum of urogenital disorders that include a glomerulopathy as a prominent feature.[168] Frasier syndrome is defined by pseudohermaphroditism and a progressive glomerular disease typically showing histologic features of FSGS.[169] Affected individuals have normal external female genitalia but an XY karyotype and streak gonads and are at high risk of the development of gonadoblastoma.[170] Many Frasier syndrome mutations in *WT1* abolish an alternative splice site that leads to an alteration of the incorporation of amino acids KTS located between two zinc fingers of the WT1 protein.[169,171–173] These mutations act in a dominant manner. Denys-Drash syndrome is a related disorder defined by pseudohermaphroditism, mesangial sclerosis, Wilms' tumor, and early progression to ESKD.[174] These disorders are perhaps best viewed as part of a phenotypic spectrum caused by *WT1* mutations, rather than distinct disorders.[170,175]

Mutations in *WT1* can also cause isolated nephrotic syndrome characterized by diffuse mesangial sclerosis or FSGS on kidney biopsy.[176] WT1 regulates the transcription of multiple genes critical to podocyte development and function.[128,168,177,178] Thus, WT1-associated glomerulopathies are presumed to be a result of altered glomerular development.[179–181] Other disorders that can be caused by WT1 defects include WAGR (***W***ilms' tumor, ***a***niridia, ***g***enitourinary anomalies, and mental ***r***etardation) syndrome, seen in individuals with a deletion in a region of chromosome 11p13 that includes WT1, and Meacham syndrome, a multisystem disorder that includes urogenital abnormalities caused by point mutations in WT1.[182,183]

Other Transcription Factors

Mutations in the *SMARCAL1* gene cause the autosomal recessive disorder Schimke immuno-osseous dysplasia, characterized by spondyloepiphyseal dysplasia, T cell immunodeficiency, and a glomerulopathy.[184] *SMARCAL1* regulates gene transcription through its effect on chromatin structure.[185] Mutations in *SMARCAL1* have been reported to cause a nontrivial fraction of childhood SRNS.[2] The *Pax2* gene (for paired box–containing 2) is a transcription factor with important roles in the developing nervous system and kidney.[186,187] Mutations in *Pax2* cause papillorenal syndrome, also known as *renal-coloboma syndrome*, characterized by both ocular and renal anomalies.[188] Dominant-acting mutations in *Pax2* have also been reported as a rare cause of isolated, late-onset FSGS.[189] A mutation in the *NXF5* transcription factor gene has been reported to cause an X-linked phenotype characterized by FSGS and heart block in one large family.[190]

OTHER GENES

Mutations in the phospholipase C epsilon 1 gene, *PLCE1*, cause an early-onset recessive form of nephrotic syndrome.[191] Typically, affected individuals develop early onset nephrosis and diffuse mesangial sclerosis (DMS). In nonsyndromic DMS, mutations in *PLCE1* have been reported in 10 of 35 families.[192] *PLCE1* mutations have also been reported as a cause of congenital nephrotic syndrome.[2] PLCE1 is a phospholipase protein. These proteins catalyze the formation of secondary messengers, including 1,4,5-trisphosphate and diacylglycerol. The mechanistic connection between PLCE1 mutations and DMS is not understood.

Mutations in the podocalyxin *PODXL* gene have been reported as a possible rare cause of autosomal dominant FSGS.[193] Podocalyxin is a member of the CD34 sialomucin protein family that is highly expressed in podocytes. Recently, an infant with congenital nephrotic syndrome, as well as omphalocele and microcoria, was found to harbor loss-of-function mutations on both PODXL alleles.[194] This recessive phenotype is similar to the phenotype in *PODXL*-deficient mice.[195]

Other genes that have been reported to cause rare recessive forms of nephrotic syndrome when mutated include epithelial membrane protein 2 (*EMP2*)[196] and the Crumbs cell polarity complex gene 2 (*CRB2*).[197,198]

Fibronectin glomerulopathy is an autosomal dominant disease characterized clinically by hematuria, proteinuria, frequent hypertension, and progressive renal failure, with variable penetrance and age of onset.[199] Renal histology is notable for extensive mesangial and subendothelial deposits that contain fibronectin and are largely devoid of immunoglobulin or complement. Mutations in FN1, the gene encoding for fibronectin, are detected in a significant proportion of families presenting with this disease.[200] Disease recurrence in kidney allografts suggests deposition of circulating fibronectin as the disease mechanism, although the pathophysiology is not well understood.[201]

COMMON GENETIC FACTORS

APOL1 NEPHROPATHY

People of recent African ancestry have a markedly increased risk of focal segmental glomerulosclerosis, as well as other forms of nondiabetic kidney disease.[5,6] This increased risk is largely attributable to two variants in the *APOL1* gene encoding apolipoprotein L1. APOL1-associated kidney disease is inherited largely as a recessive trait. These two variants alter the APOL1 amino acid sequence and its function. The first allele, known as G1, leads to two amino acid substitutions near the C-terminus that nearly always occur together, S342G and I384M. The second allele, G2, leads to a 6-bp deletion of amino acid residues 388 and 389. G1 and G2 never occur on the same chromosome. APOL1 variants increase the risk of several subtypes of kidney disease, with inheritance of two risk alleles leading to markedly increased risk. Case-control odds ratios are approximately 7 to 10 for hypertension-attributed ESKD, 10 to 17 for FSGS, and 29 to 89 for HIVAN.[5,202,203] The G1 and G2 forms of APOL1 appear to protect individuals against the trypanosomes that cause African sleeping sickness.[5] The G1 and G2 alleles have a combined allele frequency of approximately 35% in African Americans.[5] In Africa, G1 and G2 are most frequent in western sub-Saharan Africa, with much lower frequencies in eastern Africa.[204] Although a high-risk APOL1 genotype is an extremely strong risk factor for FSGS and other forms of nondiabetic kidney disease, most people with these genotypes do not develop overt kidney disease.

The product of APOL1, apolipoprotein L1 (ApoL1), is a secreted lipoprotein that circulates as part of high-density lipoprotein 3 (HDL3) complexes, the densest HDL fraction, and is expressed in multiple tissue types.[205–209] Most but not all studies have supported a model in which the G1 and G2 forms of APOL1 have gain-of-function effects that lead to increased kidney disease risk. In cells, mice, and drosophila, expression of the G1 and G2 forms of APOL1 lead to increased death of the APOL1-expressing cells.[210–216]

OTHER COMMON GENETIC FACTORS

Noncoding variants in the *GPC5* gene, encoding glyipcan-5, have been shown to associate with a risk of nephrotic syndrome. Glypican-5, part of a six-member family of related glypicans, is a heparin sulfate proteoglycan expressed in podocytes. The disease-associated allele is associated with higher levels of GPC5 expression.[217]

Variation near the HLA-DQ locus has been found to be associated with an increased risk of steroid-sensitive nephrotic syndrome by a genome-wide association study.[14] Subsequent work in steroid-sensitive nephrotic syndrome subjects of different ethnicities has supported the association with HLA-DQB1 and also have shown associations with HLA-BRB1 and near *BTNL2*. An increased burden of risk alleles across these three independent loci was found to be associated with a higher risk of nephrotic syndrome.[13]

IMMUNOGLOBULIN A NEPHROPATHY

IgA is a common form of glomerular disease with a higher incidence in Asia. Multiple families segregating IgA nephropathy have been identified and studied. Early genetic studies in families from Kentucky and Italy have suggested the existence of a major IgA locus on Chr6q22-23.[218] Additional studies have suggested genetic heterogeneity of familial IgA nephropathy, implicating other genomic loci.[219] The genes and their variants underlying these familial forms have not been identified.

Several genome-wide association studies (GWAS) have been performed in an effort to identify more common genetic variations involved in susceptibility to IgA nephropathy. Several loci have been convincingly implicated as playing important roles. Loci within the MHC locus, as well as a deletion variant in CFHR1 and CFHR3 (Chr1q32), have been implicated.[9,10] Subsequent larger GWAS have identified additional loci, including signals at genes *ITGAM-ITGAX, VAV3, CARD9*, *DEFA, ST6GAL1,* and *ODF1-KLF10*.[12,220]

The association of IgA nephropathy with an 84 kb deletion involving *CFHR1* and *CFHR3* clearly implicates the alternative complement pathway as playing a role.[10] This deletion has been shown to associate with a reduced level of mesangial immune deposits.[221] Several of the loci implicated have also been shown to be associated with an altered risk of inflammatory bowel disease and with regulating the intestinal epithelial barrier and may reflect evolutionary pressures reflecting human interaction with intestinal pathogens.[12] For the most part, causal variants at these loci have not been clearly identified, nor their causal relationship to disease clarified. GWAS studies have also identified loci in *C1GALT1* and *C1GALT1C1,* encoding genes necessary for mucin type O–linked glycosylation as affecting levels of galactose-deficient IgA1.[222,223] Galactose-deficient IgA1 levels are higher in patients with IgA nephropathy compared with healthy controls[224] and are associated with disease progression in IgA but not membranous nephropathy.[225]

MEMBRANOUS NEPHROPATHY

Membranous nephropathy (MN) is discussed at length in Chapter 31. Highly penetrant gene mutations causing monogenic forms of membranous nephropathy have not been reported. However, MN has significant genetic components. GWAS have identified alleles that are highly associated with the risk of idiopathic MN at two genomic loci. Variation at the M-type phospholipase A2 receptor—PLA(2)R1—gene on Chr2q24 is highly associated with MN, consistent with an alteration in the PLA(2)R antigen, which is the major target antigen in this disorder. In addition, variations at HLA-DQA1 are also highly associated with a risk of MN, in addition to risks of other forms of immune-mediated glomerulonephritis[7,8]

DISEASES OF THE GLOMERULAR BASEMENT MEMBRANE

The GBM is one of the three major structural components that compromise the filtration barrier of the kidney. It is a fibrous extracellular matrix positioned between the glomerular podocyte on one side and the capillary endothelium and mesangium on the other. Ultrastructurally, the membrane is composed of three layers, a central lamina densa sandwiched between the lamina rara interna (on the endothelial side) and lamina rara externa (on the podocyte side). The GBM is composed of dozens of matrix proteins, the most prominent of which are type IV collagen, laminins, the heparin sulfate proteoglycan agrin, and nidogen.[226–229] During development, the immature GBM forms from the fusion of basement membrane laid down by both capillary endothelial cells and podocytes, with both cell types contributing to the formation and later maintenance of the mature GBM.[230–232] The maturation of the GBM is characterized by a significant change in its molecular composition; the laminin composition switches from laminin $\alpha_1\beta_1\gamma_1$ (LM-111) and laminin $\alpha_5\beta_1\gamma_1$ (LM-511) in the developing GBM to laminin $\alpha_5\beta_2\gamma_1$ (LM-521) in the mature glomerulus, whereas collage IV expression changes from collagen IV $\alpha_1\alpha_2\alpha_1$ (IV) to collagen IV $\alpha_3\alpha_4\alpha_5$ (IV). Additional changes in the molecular composition of the GBM take place during glomerular development and await more systematic study. Its origin as a fusion of matrix laid down by two distinct cells on opposite sides of the GBM may explain why its thickness (250–400 nm, depending on the study[233,234]) is relatively large for a basement membrane.

The GBM is thought to play several critical roles in the function of the filtration barrier. Beyond providing a structural scaffold capable of resisting transglomerular pressure, matrix components act as adhesion and signaling receptors for podocytes and endothelial and mesangial cells. In addition, the matrix is thought to act as a potential reservoir for soluble signaling molecules.[235] Morphologic and molecular changes to the GBM occur in numerous human diseases and various animal models, with diabetic nephropathy being the classic example. The role of the GBM as an actual filter in maintaining the permselectivity of the glomerulus remains an unresolved issue. Farquhar and colleagues and others have long argued that the biophysical properties of the matrix, including its electrostatic charge, play a critical role in the size and charge selectivity of the glomerular filtration barrier.[236–240] Studies by other groups have variously suggested that the endothelium and inner aspect of the GBM are the major barriers to macromolecules,[241,242] the slit diaphragm is the crucial filter,[243–245] and the GBM and slit diaphragm form a double barrier.[246–248] Detailed analysis of Lamb2-deficient mice (a model of Pierson syndrome) has demonstrated that albuminuria precedes foot process effacement in these animals,[249] supporting the notion that a properly assembled GBM functions as at least a crude prefilter. The critical importance of the GBM in maintaining glomerular structure and function is most vividly demonstrated by humans carrying mutations in the *COL4A3*, *-4,* or *-5* genes (and presenting with Alport syndrome or thin basement membrane nephropathy) or the *LAMB2* gene (and presenting with Pierson syndrome or congenital nephrotic syndrome).

ALPORT SYNDROME AND COLLAGEN IV–RELATED RENAL DISEASE

The clinical syndrome of hereditary nephritis and associated hearing loss, with more severe disease in males but inheritance through females, was described by Alport in 1927.[250] Williamson, who described several pedigrees with hematuria, renal failure, and variable deafness and eye defects, proposed naming the syndrome after Alport in 1961.[251] Mutations in any one of three type IV collagen genes (*COL4A3*, *COL4A4*, and *COL4A5*) are now known to be responsible for both Alport syndrome and benign familial hematuria–thin basement nephropathy.[252] Between 30,000 and 60,000 people have been estimated to be affected by Alport syndrome in the United States,[253] accounting for 0.3% of incident ESKD patients.[254] However, more recent genetic studies have suggested that mutations in *COL4A3* and *COL4A4* may account for a significant number of cases of renal disease that have not been clinically or pathologically diagnosed as Alport syndrome or thin basement membrane disease.[255–259]

Type IV collagens are a major component of basement membranes and are composed of alpha chains encoded by six genes (*COL4A1–COL4A6*), comprising only a fraction of the different types of collagens found in vertebrates. In humans, the six collagen IV genes are arranged as three pairs of head to head genes (*COL4A1* and *COL4A2* on chromosome 13, *COL4A3* and *COL4A4* on chromosome, and *COL4A5* and *COL4A6* on the X chromosome). All six chains share a common protein structure, comprised of an amino terminal domain rich in cysteines and lysines, termed the *7S domain,* the triple helical collagenous domain, made up of several hundred Gly-X-Y repeats, and finally a carboxyl terminal noncollagenous (NC1) domain. The collagenous domains of type IV collagen alpha chains are notable for between 21 and 26 interruptions of the Gly-X-Y repeats, which provide type IV collagens with the flexibility necessary to form networks; the interruptions may also act as specific interchain cros-slinking and cell adhesion sites.[260] Type IV collagen network formation occurs through a series of distinct steps (Fig. 43.2). First, three alpha chairs form a trimeric protomer with all three chains in parallel; this assembly occurs in the secretory pathway of the cell. Once secreted, these protomers can then form higher order complexes via tetrameric head to head interactions of their 7S domain and dimeric tail to tail binding of the NC1 domains.[261] These structures are further stabilized through supramolecular twisting, lysyl oxidase–mediated cros-slinks, and disulfide and sulfilimine bonds.[262,263] Ultimately, the type IV collagens generate an irregular polygonal network in the GBM[264] with which other extracellular matrix (ECM) components can interact.

Although there are six distinct collagen IV alpha chains, only three stoichiometries of protomers—$\alpha_1\alpha_1\alpha_2$ (IV), $\alpha_3\alpha_4\alpha_5$ (IV), and $\alpha_5\alpha_5\alpha_6$ (IV)—are found. At the NC1 domain end, $\alpha_3\alpha_4\alpha_5$ (IV) protomers can only homodimerize, whereas trimerized $\alpha_1\alpha_1\alpha_2$ (IV) NC1 domains can bind either other $\alpha_1\alpha_1\alpha_2$ (IV) NC1 trimers or $\alpha_5\alpha_5\alpha_6$ (IV) NC1 trimers. This gives rise to only three potential hexamer-generated networks: $\alpha_1\alpha_1\alpha_2$ (IV)–$\alpha_1\alpha_1\alpha_2$ (IV), $\alpha_3\alpha_4\alpha_5$ (IV)–$\alpha_3\alpha_4\alpha_5$ (IV), and $\alpha_1\alpha_1\alpha_2$ (IV)–$\alpha_5\alpha_5\alpha_6$ (IV).[265–267] A defect in one component of a protomer due to a gene mutation prevents the normal expression of the other components in the network; for example, a homozygous mutation in *COL4A3* will prevent the expression of COL4A4, whereas COL4A5 will be present only when incorporated into the $\alpha_5\alpha_5\alpha_6$ (IV) network.

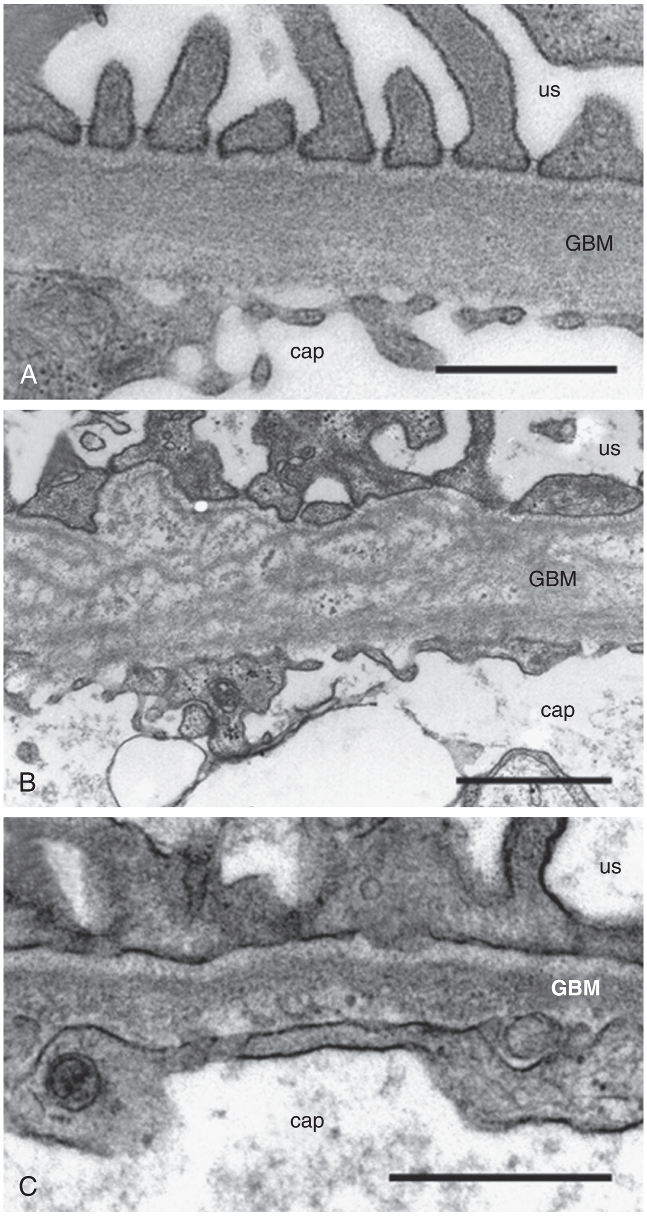

Fig. 43.2 Transmission electron microscopy appearance of the glomerular basement membrane *(GBM)* in normal conditions, Alport syndrome, and thin basement membrane nephropathy (TBMN). (A) In the normal glomerulus, GBM width is uniform and from 250 to 400 nm in size. (B) In Alport syndrome, focal thickening and thinning of the GBM is observed with the development of a laminated GBM and a woven basket appearance. (C) In TBMN, the GBM is characteristically thinned; its thickness is only about half that of a normal kidney. *Bars* represent 500 nm. *CAP,* Capillary lumen; *US,* urinary space. (A, Courtesy Dr. Finn P. Reinholt, Karolinska University Hospital, Huddinge, Sweden; B, courtesy Dr. Kjell Hultenby, Karolinska University Hospital, Huddinge, Sweden.)

In the kidney, collagen IV $\alpha_1\alpha_1\alpha_2$ is present in the GBM, mesangium, Bowman's capsule, and tubular basement membranes. Collagen IV $\alpha_3\alpha_4\alpha_5$ expression is switched on in podocytes during glomerular maturation and largely supplants $\alpha_1\alpha_1\alpha_2$ (IV) in the GBM[230,268]; it is also found in some tubular basement membranes. Outside the kidney, $\alpha_3\alpha_4\alpha_5$ (IV) networks are also found in the cochlea, eye, lung, and testis.[269–272] The $\alpha_5\alpha_5\alpha_6$ (IV) network is present in the basement membrane of Bowman's capsule and tubules but not in the GBM,[273] a distinction that has been used to differentiate Alport syndrome driven by *COL4A5* versus *COL4A3* and *COL4A4* mutations.[274,275]

The precise mechanism whereby the absence of a functional collagen $\alpha_3\alpha_4\alpha_5$(IV) network in Alport patients leads to abnormal GBM architecture and ultimately glomerular failure remains to be established, but several pathways have been implicated. The GBM in Alport disease demonstrates persistence of the collagen $\alpha_1\alpha_1\alpha_2$ (IV) network in the GBM, as well as deposition of type V and VI collagens and laminin α_2.[268,276] The $\alpha_1\alpha_1\alpha_2$ (IV) network is less cross-linked than the $\alpha_3\alpha_4\alpha_5$ (IV) network and therefore may be more susceptible to endoproteolysis. In addition, the $\alpha_1\alpha_1\alpha_2$ (IV) network may provide inappropriate signals to podocytes via the discoidin domain receptor 1 (DDR1), integrin α_1, and integrin α_2, as judged by studies in murine models.[277–280] Aberrantly expressed laminin α_2 may also stimulate pathogenic signaling and interfere with the function of LM-521.[281,282]

Animal and human studies have also suggested a strong pathologic interplay between loss of the $\alpha_3\alpha_4\alpha_5$ (IV) network and the hemodynamic pressure across the glomerular filtration barrier. Inducing hypertension substantially aggravates disease in Alport mice,[283] whereas ACE inhibitor treatment prolongs renal survival in both mice and humans.[284–286] Accelerated detachment of podocytes from the GBM during Alport syndrome progression is likely involved,[286–288] as may be endothelin-A mediated endothelial-mesangial cross-talk,[289] a potential therapeutic target.

GENETICS OF ALPORT SYNDROME AND COLLAGEN IV–ASSOCIATED RENAL DISEASE

Alport syndrome can be caused by mutations in *COL4A3*, *COL4A4,* and/or *COL4A5*.[290–293] *COL4A5* is located on the X chromosome and is responsible for X-linked forms of Alport syndrome, whereas mutations in *COL4A3* and *COL4A4* (both located on chromosome 2) are responsible for both autosomal recessive (AR) and autosomal dominant (AD) forms of Alport syndrome. In addition, rare digenic forms of Alport syndrome have been reported, involving either combined *COL4A3-COLA4* mutations in trans (mimicking AR disease), combined *COL4A3-COLA4* mutations in cis (mimicking AD disease), and mutations in *COL4A5* combined with *COL4A3* or *COL4A4* mutations.[294] Many dozens of distinct mutations, including deletions, frameshifts, nonsense, and missense mutations, have been described for each of these genes. Family-based studies have reported that X-linked disease accounts for about 80% of cases, with 15% showing AR inheritance and the remaining 5% demonstrating AD inheritance. However, more recent results using next-generation sequencing technology have revealed a higher incidence (20%–30%) of AD forms.[295,296]

Males carry only one X chromosome and hence will develop X-linked Alport syndrome if they carry a mutation in the *COL4A5* gene.[290] Affected males invariably experience microscopic hematuria, subnephrotic proteinuria, and progressive renal disease, usually culminating in ESKD. The rate of disease progression and extrarenal manifestations, including sensorineural deafness and ocular findings, are strongly influenced by the underlying mutation, with large deletions and nonsense, splice site, and frameshift mutations portending a worse prognosis relative to missense mutations.[297,298] The occasional association with leiomyomatosis has been attributed to coexisting mutations in the adjacent *COL4A6* gene.[299,300]

Female carriers demonstrate mosaic expression of $\alpha_3\alpha_4\alpha_5$ (IV) networks within glomeruli due to random X-linked inactivation. Although in mouse models of *COL4A5* disease, skewing of inactivation of the disease-carrying X chromosome has been correlated with disease severity in female heterozygotes,[301] data for humans are less clear, with large intrafamilial variability and no clear genotype-phenotype correlation.[302] In contrast, the development of proteinuria and hearing loss do portend a poorer renal prognosis in these carriers.[302]

AR Alport disease has been described with mutations in both *COL4A3* and *COL4A4*.[292,293] Males and females are affected equally, and the clinical presentation is similar to males with *COL4A5* mutations, with frequent sensorineural hearing loss and ocular abnormalities.[303,304] AD forms of Alport syndrome due to mutations in *COL4A3* or *COL4A4* have also been reported.[305] In general, patients with AD disease are less likely to have an ocular phenotype and tend to demonstrate slower progression of renal failure.[306–308] However, due to the broad spectrum of phenotypes associated with heterozygous mutations in these two genes,[309] differentiating between AD Alport disease and thin basement membrane nephropathy can be difficult and impractical (see later).

Thin basement membrane nephropathy is a pathologic diagnosis associated with glomerular hematuria and the finding of relatively thin GBMs on transmission electron microscopic examination. It can be seen in a variety of scenarios, including patients destined to develop Alport syndrome (e.g., young patients with AR Alport syndrome and young males with *COL4A5* mutations), female carriers of *COL4A5* mutations, persons with heterozygous *COL4A3* or *COL4A4* mutations, and patients with mutations in other genes or unknown genetics.[225,252,310–313]

Traditionally, renal disease associated with *COL4A3, COL4A4,* and *COL4A5* gene mutations have been divided into Alport syndrome and thin basement membrane nephropathy–benign familial hematuria based on clinical and histologic criteria. However, with the appreciation that thin basement membrane disease carries a significant risk of progression to overt CKD and even ESKD, and with the finding of heterozygous *COL4A* gene mutations in a significant proportion of patients clinically diagnosed as FSGS,[255–259] several groups, including the Alport Syndrome Classification Working Group, have suggested a new classification driven by the genetic underpinning of the disease.[309,314] The utility of such a scheme will largely depend on how widely available genetic testing becomes, but could be of particular value for patients and families with heterozygous *COL4A3* or *COL4A4* mutations. In the absence of routine genetic testing, it is important that clinicians and pathologists be cognizant of the possibility of collagen IV–associated disease as the underlying pathology, even in the absence of classic clinicopathologic findings.

CLINICOPATHOLOGIC PRESENTATION OF ALPORT SYNDROME AND COLLAGEN IV–RELATED KIDNEY DISEASE

Clinically, Alport syndrome is characterized by glomerular hematuria, subnephrotic range proteinuria, and progressive renal failure associated with sensorineural hearing loss.[250,251] For patients with X-linked or AR Alport syndrome, microscopic hematuria is nearly universal by early childhood. Although the time to reach ESKD is variable, cohort studies have reported similar median renal survival in the mid-20s for X-linked cases[297] and AR disease.[303,304] Hearing loss is detected in more than half of these patients; it begins with loss of high tones but frequently progresses to impair conversational communication. Eye defects can include anterior lenticonus, perimacular retinal flecks, and corneal erosions and are quite frequent but only rarely impair vision.[315] Leiomyomatosis has been reported to affect a subset of patients with X-linked disease and is attributed to concomitant *COL4A6* mutations.[299,300,316]

Female carriers of *COL4A5* mutations and persons with heterozygous *COL4A3* or *COL4A4* mutations can develop a variety of findings, from isolated microscopic hematuria to gross hematuria, proteinuria, renal dysfunction, and sensorineural hearing loss. In line with this, other studies have suggested that carriers of *COL4A3* or *COL4A4* mutations and female carriers of *COL4A5* mutations, who may have initially been diagnosed with benign familial hematuria or thin basement membrane disease, have a significantly higher risk of progression to CKD and ESKD than previously suspected. Among *COL4A5* carriers, 12% progressed to ESKD by age 40 years in a large cohort study, with proteinuria and hearing loss identified as risk factors.[302] Among patients with heterozygous *COL4A3* or *COL4A4* mutations, the reported risk of progressing to ESKD is variable and probably reflects ascertainment bias of familial disease, as well as significant variability in the pathogenicity of individual mutations. One study of AD Alport syndrome reported a median renal survival of only 31 years,[308] another a median renal survival of 51 years,[306] and a third study found a lower rate and later onset of ESKD.[317] Reported risk factors for ESKD include proteinuria, hearing loss, progressive decline in GFR in the patient or family members, and biopsy findings of FSGS, GBM thickening, or GBM lamellation.[314]

Prior to the availability of genetic testing, the diagnosis of Alport syndrome was confirmed by renal pathology or skin biopsy.[275,318,319] By light microscopy, glomerular morphology can be relatively normal prior to the age of 5 years, after which mesangial cell proliferation and matrix accumulation, as well as capillary wall thickening, may be seen. This progresses to glomerular sclerosis and associated tubular and interstitial pathology over time.[320–322] Standard immunofluorescence microscopy findings are nonspecific. However, staining for collagens IV α_3, α_4, and α_5 demonstrate an absence of all three chains in the GBM in the large majority of X-linked and many AR Alport cases.[271,274,275,323,324] In contrast, female *COL4A5* mutation carriers can demonstrate segmental loss of staining in skin and GBM due to random X-linked inactivation of the wild-type allele. By transmission electron microscopy, early Alport syndrome or thin basement membrane disease can demonstrate thinning of the GBM.[325,326] In more advanced disease, the classic findings of GBM lamellation and a woven basket appearance, as well as irregular thinning and thickening of the GBM, are seen (Fig. 43.3).[327] These electron microscopy findings, when paired with collagen IV immunofluorescence staining, are highly suggestive of Alport disease. Other genetic studies have demonstrated collagen IV gene mutations in a significant percentage of patients carrying diagnoses other than Alport syndrome or thin basement membrane nephropathy,[255–259] suggesting that the sensitivity of clinical and pathologic findings may be lower than previously appreciated for identifying collagen IV–associated renal disease.

TREATMENT OF ALPORT SYNDROME

Treatment options for Alport syndrome are currently limited to the use of ACE inhibitors. Based on mouse studies,[284] ACE inhibitors and ARBs have been tested in Alport patients and have shown efficacy in reducing proteinuria.[328–330] Treatment with ACE inhibitors is associated with a significant delay in progression to ESKD and median life expectancy, with early treatment initiation providing the greatest benefit.[285] The benefit of ACE inhibitors has been extended to patients with heterozygous mutations or carriers of X-linked and AR Alport mutations.[331,332] Some experts now recommend treatment for anyone with an Alport disease gene mutation, hematuria, and either overt proteinuria or microscopic albuminuria and a severe mutation (e.g., deletion, nonsense, or splicing), hearing loss or family history of ESKD prior to the age of 30 years.[314]

Multiple treatment strategies have been explored in animal models of Alport syndrome. These include inhibition of TGF-β signaling,[277] treatment with an endothelin A antagonist,[289] and inhibition of microRNA-21 with the use of an anti-miR.[333] Clinical trials are underway to examine the potential benefit of an micro-RNA inhibitor and bardoxolone methyl in patients with Alport syndrome.

Kidney transplantation should be considered for Alport patients who progress to ESKD. Overall, patient and graft outcomes are good, although patients with severe mutations are at risk for developing de novo anti-GBM nephritis. This is due to the presence of collagen $\alpha_3\alpha_4\alpha_5$ (IV) in the allograft, which is absent in the host. Antibodies against these collagens in the serum and linear immunoglobulin deposits along the GBM are detectable in a significant proportion of allografts, but are often not associated with evidence of nephritis.[334] The reported incidence of anti-GBM nephritis in Alport patients status posttransplantation is variable, but most series have reported less than 3% of patients affected.[335–339]

PIERSON SYNDROME

A familial form of congenital nephrotic syndrome associated with a complex eye phenotype, including microcoria (extremely narrow, nonreactive pupils) was first described by Pierson and associates in 1963.[340] Although similar cases were described in the literature in the antecedent decades, it was not until 2004 that the term *microcoria-congenital nephrotic syndrome* or *Pierson syndrome* was proposed to describe this autosomal recessive disorder.[341] Mutations in *LAMB2*, encoding for the beta-2 subunit of laminin, are responsible for the disease[342]; many aspects of the pathology are seen in *Lamb2*-deficient mice.[343,344]

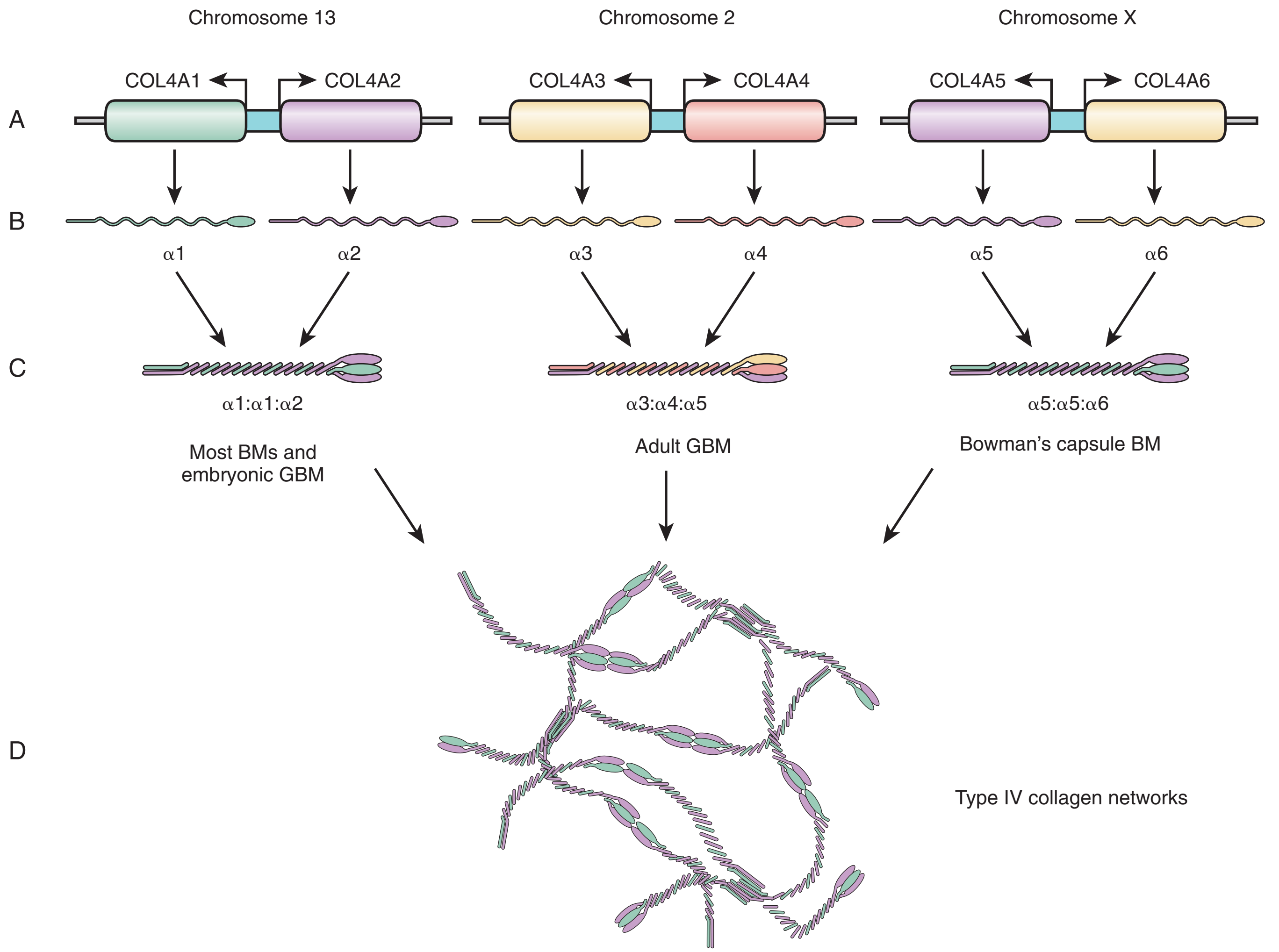

Fig. 43.3 Type IV collagen genes, alpha chains, and glomerular isoforms. (A) The six collagen IV genes (*COL4A1* to *COL4A6*) are located pairwise in a head to head manner on three different chromosomes. (B) The *COL4A* genes generate six different alpha chains that have a globular noncollagenous domain at their C terminus. (C) Three chains form three combinations of triple-helix molecules. (D) Extracellularly, the triple-helix type IV collagen molecules form a network by associating with each other at their ends so that two molecules are cross-linked through their C-terminal globular domain (NC1), and four trimers associate with each other at the N-terminal 7S domain. The ubiquitous α_1:α_1:α_2 trimer is the only isoform in the embryonic glomerular basement membrane *(GBM)*. After birth, this isoform is gradually replaced by the α_3:α_4:α_5 isoform. Defects in the α_3:α_4:α_5 trimers lead to a variety of glomerular pathology, ranging from thin basement membrane nephropathy to Alport syndrome (see text).

Laminins are trimeric, cross-shaped matrix proteins assembled out of one alpha, one beta, and one gamma chain. In the GBM, the major laminin isoform is laminin-521 (LM-521), consisting of the α_5, β_2, and γ_1 subunits.[345] Trimers further polymerize to form a lattice network,[346] which can anchor additional GBM components, including agrin and nidogen. Laminins act as cell adhesion receptors, with the carboxyl terminal laminin globular (LG) domain of the alpha chain a major site for integrin and dystroglycan binding.[347] LM-521 is secreted by both endothelial cells and podocytes.[231]

Pierson syndrome is an autosomal recessive disease. Both compound heterozygous and homozygous mutations have been reported. Most mutations reported are severe truncating or frameshift mutations and appear to completely abolish laminin beta-2 protein in the GBM. Even truncation mutations smaller than 50 amino acids from the carboxyl terminus of LAMB2 appear to be functional null alleles. Missense mutations have been reported, with evidence of partial LM-521 function and milder disease, including childhood presentation of nephrotic syndrome and later development of ESKD.[348–351] Heterozygous missense and promoter variant single-nucleotide polymorphisms (SNPs) have been reported in patients with nephrotic syndrome, but their clinical significance remains uncertain.[351] Clear correlations between renal and nonrenal phenotypes and between genotype and neurologic phenotypes are lacking.

Clinically, most patients present at or shortly after birth with the symptoms of congenital nephrotic syndrome in addition to a variety of nonrenal deficits.[341,351,352] Cases of nephrotic syndrome presenting in childhood have been reported but are uncommon; most patients progress rapidly to ESKD.[351] Isolated nephrotic syndrome is rarely caused by

LAMB2 mutations.[24,351] Ocular defects are almost universal, with microcoria, lenticonus, or cataracts and retinal detachments most common.[351,353,354] Ocular and/or neurodevelopmental defects due to *LAMB2* without renal disease has not been described, although rare cases of ocular disease preceding the development of nephrotic syndrome have been reported. Additional neurodevelopmental defects have also been reported, including muscular hypotonia and progressive neurocognitive involvement.[340,342,351] Diffuse mesangial sclerosis with associated diffuse podocyte foot process effacement is seen on renal biopsy.

Treatment for Pierson syndrome is currently limited to supportive therapy. Successful renal transplantation has been reported, with improvement in renal failure associated with growth retardation.[355] However, the presence of significant neurodevelopmental defects may limit the utility for this approach in other cases. Interestingly, recombinant LM-521 trimers infused into *Lamb2*-deficient mice appear to accumulate in the GBM and delay the development of severe proteinuria.[356] However, the LM-521 trimers appear to accumulate mainly on the endothelial half of the GBM, and a beneficial effect of the treatment on renal survival was not reported. Nonetheless, these early preclinical studies have suggested the possibility of affecting GBM composition and perhaps function through the administration of exogenous matrix components.

LAMA5

LAMA5 encodes for the laminin α_5 subunit. *LAMA5* variants have been reported in several patients with FSGS in conjunction with additional gene mutations[256,357,358] and in homozygous form in three families with apparent recessive monogenic nephrotic syndrome.[359] In the mouse, *Lama5* deletion results in severe developmental defects, a hypomorphic allele develops a proteinuria, hematuria, and polycystic kidney disease (PKD) phenotype, and podocyte-specific *Lama5* deletion produces GBM defects and proteinuria.[360–362] However, the clinical significance of *LAMA5* variants in the development of human glomerular disease remains uncertain.

SYSTEMIC DISEASES WITH GLOMERULAR COMPONENTS

Here we briefly review systemic monogenic diseases with glomerular components. This distinction is somewhat arbitrary because many of the forms of inherited glomerular diseases discussed previously have extrarenal components.

FABRY DISEASE

Fabry disease is a rare X-linked disease with both serious extrarenal and renal manifestations.[363,364] It is caused by mutations in the alpha-galactosidase gene *GLA*.[363] Dermatologic manifestations include a neuropathy characterized by reduced sensation in the palms and soles, anhidrosis, and frequent angiokeratomas. Cardiac abnormalities and neurologic symptoms are common. At a biochemical level, disease is caused by a deficiency of alpha-galactosidase activity, leading to an accumulation of globotriaosylceramide (GL3) in blood vessels and tissues, including podocytes, endothelial cells, epithelial cells, and tubular cells within the kidney. Fabry disease is treated with administration of recombinant alpha-galactosidase.[365] Treatment appears to be effective in clearing GL3 deposits from the kidney and improving the kidney prognosis.[366] Other small molecule–based therapeutic options are under investigation.[367] Because of the X-linked nature of the disease, disease manifestations are highly variable in women with GLA mutations. Kidney transplantation is a viable option for patients with Fabry disease. Kidney survival in Fabry disease is as good as those receiving transplants for other causes of ESKD.[368]

GALLOWAY-MOWAT SYNDROME

Galloway-Mowat syndrome is a rare, multisystem, AR disorder. Affected individuals exhibit microcephaly, severely altered central nervous system development and, often, nephrotic syndrome.[369] *WDR73* was the first Galloway-Mowat gene to be identified.[370,371] *WDR73* encodes a WD40 repeat domain–containing protein and is thought to play a role in microtubule function and microtubule-dependent cell cycle and cell proliferative functions. Mutations in the KEOPS complex genes have been recently reported also to cause Galloway-Mowat syndrome.[372] Recessive mutations in the genes *OSGEP, TP53RK, TPRKB*, and *LAGE3* encode the four subunits that comprise the KEOPS complex (for conserved kinase, endopeptidase, and other small proteins). Mutations in these genes were reported in 37 individuals from 32 families.

NONGLOMERULAR DISEASES APPEARING AS PRIMARY GLOMERULOPATHIES

Increasingly, genetic studies are identifying variants in nonglomerular disease genes as the likely culprit in cases of apparent primary glomerular disease. There have been several reported cases of mutations in the Dent disease genes as the cause of kidney disease presenting as proteinuria or FSGS.[373–375] Similarly, mutations in nephronophthisis genes have been reported to present initially as glomerulopathies.[376–378]

GENETIC TESTING CONSIDERATIONS

Clinical genetic testing for gene alterations underlying glomerular disease is available but is not in widespread use. A variety of academic medical centers, as well as commercial clinical laboratories, offer Clinical Laboratory Improvement Amendments (CLIA)–approved genetic testing for specific glomerular disease genes of interest, as well as panels of glomerular and/or kidney disease genes. The available test centers continue to change rapidly, and the Internet may provide the most current resource in identifying appropriate testing laboratories.

A few general considerations should be kept in mind when doing genetic testing in the evaluation and management of a patient with glomerular disease:

1. The prior probability that a gene variant is causally related to disease is related to multiple factors, including age of disease onset and clinical phenotype.
   Determination of a likely genetic cause of FSGS or nephrotic syndrome becomes less likely with increasing age. Although the likelihood of finding a genetic cause underlying congenital nephrotic syndrome in a neonate is high, it is much less likely in an adult with FSGS.[3,24,379]

2. Every gene has multiple protein sequence-altering variants. The identification of a DNA change from the most common DNA sequence seen in the human population does not mean that it has a phenotype consequence. If the prior probability of having a genetic cause of disease is high, then the meaning of the test result changes. For example, the finding of an amino acid–changing variant in a patient with FSGS and/or a strong family history of FSGS, in which this same variant is present in all the affected family members, is highly likely to be causally related to disease. However, the incidental finding of, for example, an *INF2* or *TRPC6* variant in an individual with no kidney disease, is much more difficult to interpret and, without additional information about the variant or the individual, likely has no bearing on the individual's kidney status. Such variants are often referred to as variants of uncertain significance (VUS).[380] Just like incidental findings in nongenetic testing, incidental genetic test results can create information of uncertain meaning.
3. Apparent primary glomerular disease can reflect nonglomerular gene alterations producing a phenotype that masquerades as a primary glomerular disease.
   This suggests that in the clinical setting, a clinician may want to test for a broad range of genes. These might include type 4 collagen genes or *CLCN5* in the case of familial FSGS. Finding the right balance between testing for only the most obvious candidate genes and testing for variations in a long list of possible genes requires some judgment. Interpretation of the results may benefit from the involvement of a clinical geneticist, genetic counselor, or nephrologist experienced in inherited disease.

As treatments for glomerular diseases improve, making an accurate genetic diagnosis may take on greater importance, although the clinical benefit of such testing has not been demonstrated, except for some exceptional cases.[94]

Complete reference list available at ExpertConsult.com.

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